Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits

1. INTRODUCTION

1.1 Project Overview

The food delivery industry has experienced rapid growth due to increasing smartphone usage, internet accessibility, and changing consumer lifestyles. Online food ordering platforms allow customers to conveniently browse restaurants, compare prices, place orders, and track deliveries in real time. This project analyzes food ordering behaviour and consumer trends using data visualization techniques. By examining customer preferences, ordering frequency, spending patterns, ratings, payment methods, and delivery experiences, the project identifies key factors influencing consumer decisions. Tableau dashboards are used to present meaningful insights that can help restaurants and food delivery platforms improve customer satisfaction and business performance.

1.2 Purpose

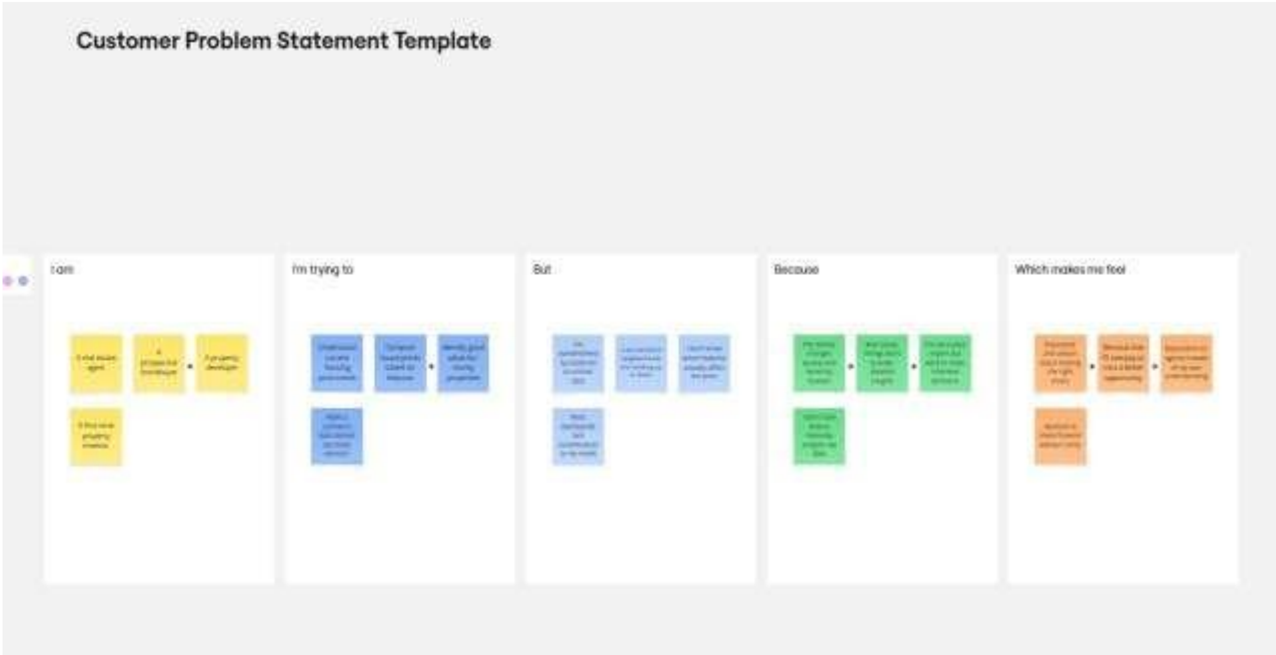
The main purpose of this project is to analyze customer food ordering behaviour and identify patterns in consumer preferences. The objectives are:

- To understand customer ordering habits.
- To identify popular cuisines and food categories.
- To analyze spending patterns and payment preferences.
- To evaluate customer satisfaction using ratings and reviews.
- To create interactive Tableau dashboards for data visualization.
- To provide insights that support better business decisions.

2. IDEATION PHASE

2.1 Problem Statement

The rapid growth of online food delivery platforms has significantly changed the way consumers order food. However, customer preferences, ordering frequency, spending habits, delivery expectations, and satisfaction levels vary across different users. Restaurants and food delivery platforms often find it difficult to identify these changing trends and make informed business decisions. Therefore, there is a need for a structured analysis of food ordering behaviour using data visualization techniques to understand consumer choices and improve customer satisfaction.



Example:

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A Busy working professional	Order healthy food quickly	I can't easily identify healthy and affordable options	Food apps prioritize promotions over nutrition	Frustrated and confused
PS-2	A college student	Order food within my budget	Prices, delivery charges, and offers change frequently	Total cost is unclear until checkout	Disappointed and hesitant

2.2 Empathy Map Canvas

An Empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes. Empathy mapping was conducted to understand the needs of stakeholders:

Says

- I want food delivered quickly.
- I prefer affordable meals.
- I like discounts and special offers.
- I expect good food quality.

- ### Thinks

  - Which restaurant has the best ratings?
  - Is the delivery charge reasonable?
  - Will my order arrive on time?
  - Is the food worth the price?

Does

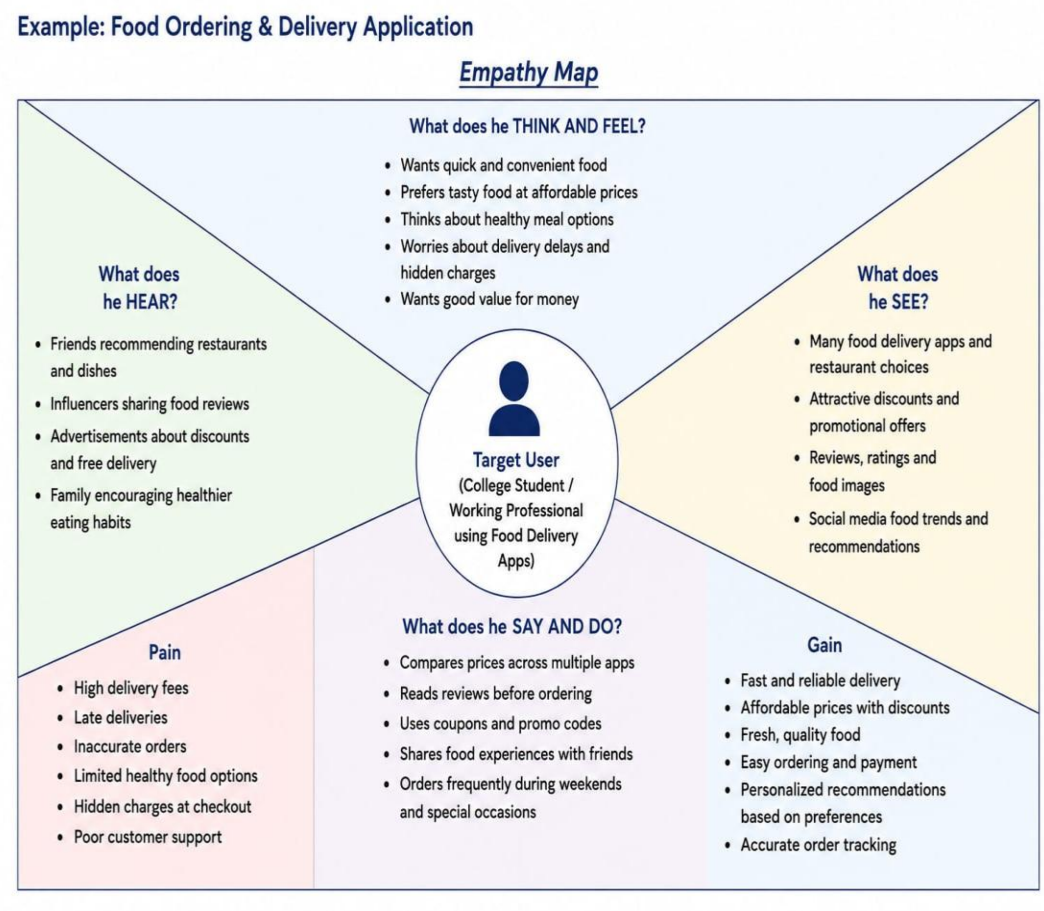
- Compares restaurants.
- Reads customer reviews.
- Applies coupons before ordering.
- Places orders through food delivery apps.

Feels

- Happy when food arrives quickly.
- Excited about discounts.
- Frustrated by delayed deliveries.
- Disappointed with poor food quality.

Example-1:

Example-2

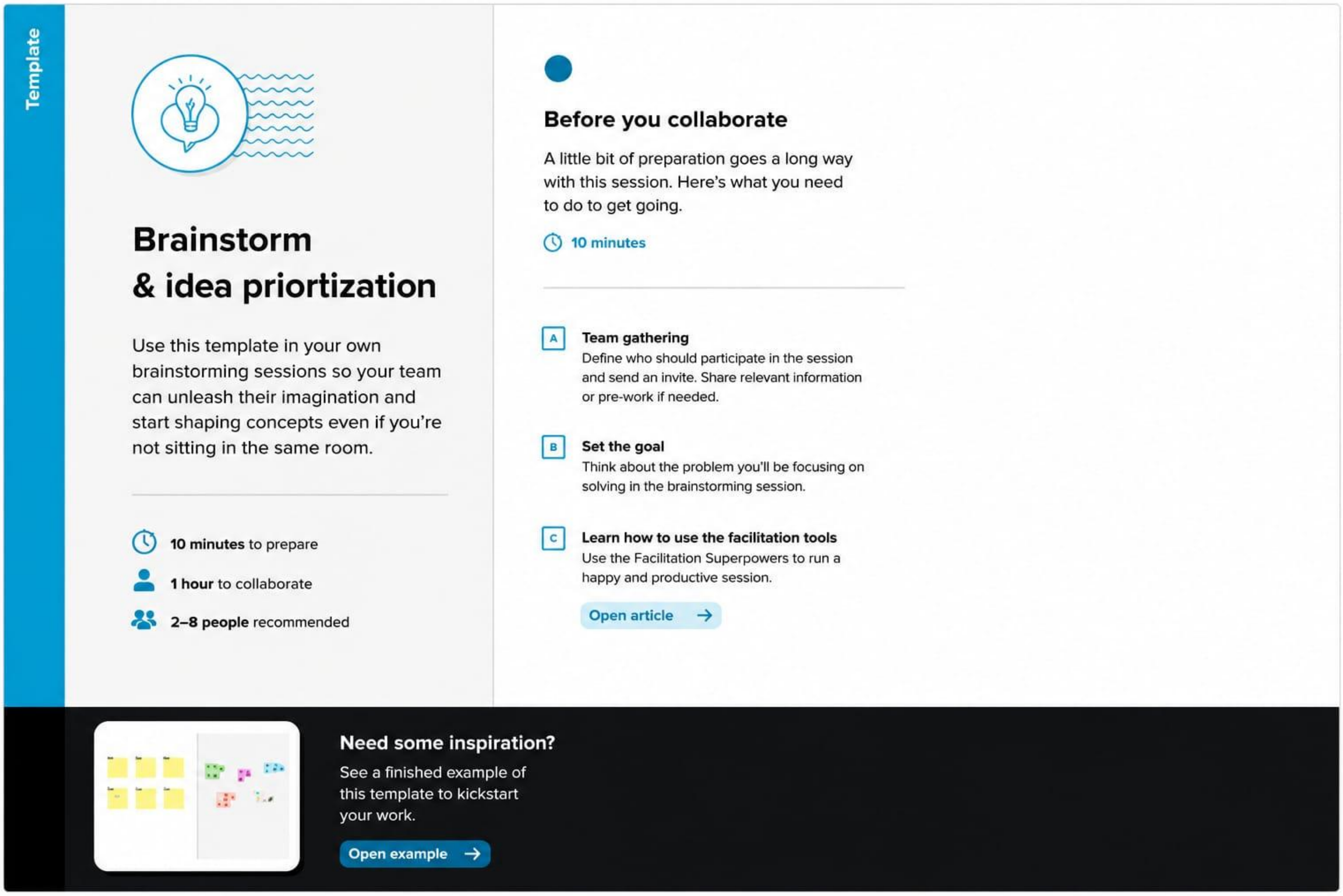


2.3 Brainstorming

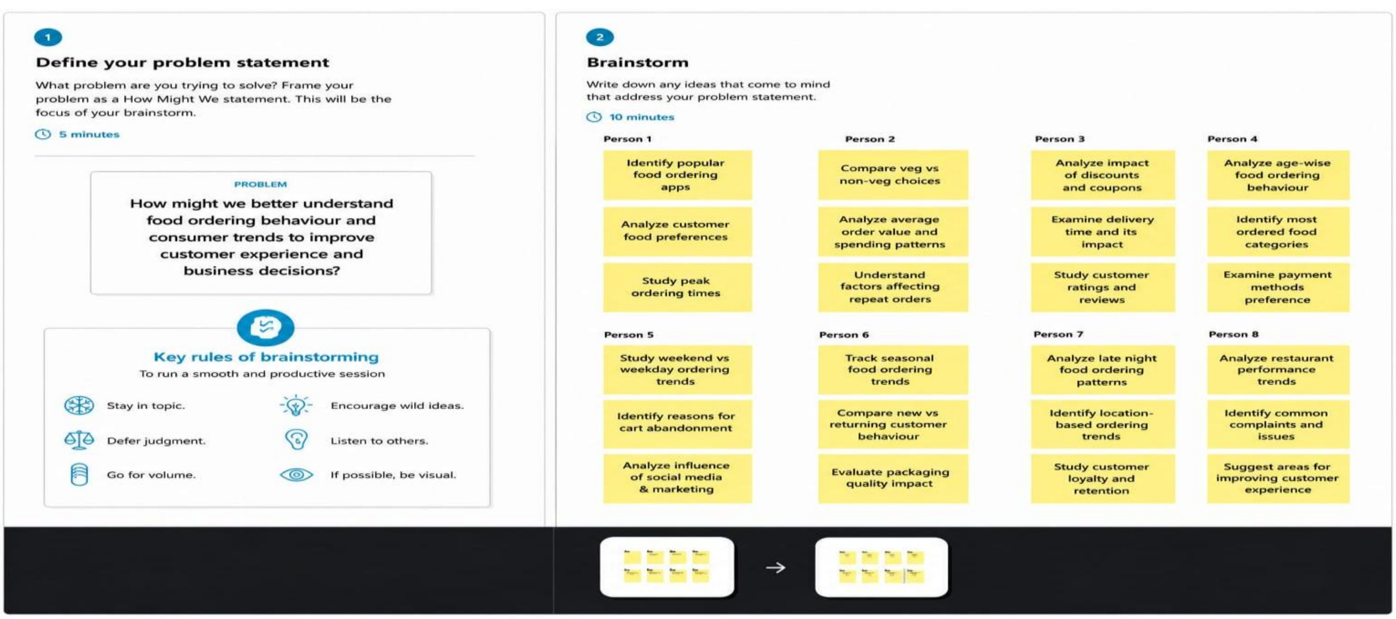
Brainstorming is a creative idea-generation technique used to identify problems, analyze customer needs, and develop effective solutions. In this project, brainstorming was conducted to understand the factors that influence food ordering behaviour and consumer preferences in online food delivery platforms.

- Create interactive Tableau dashboards.
- Analyze ordering frequency.
- Study payment method preferences.
- Evaluate customer ratings and reviews.
- Identify peak ordering hours.
- Recommend business improvements based on insights.

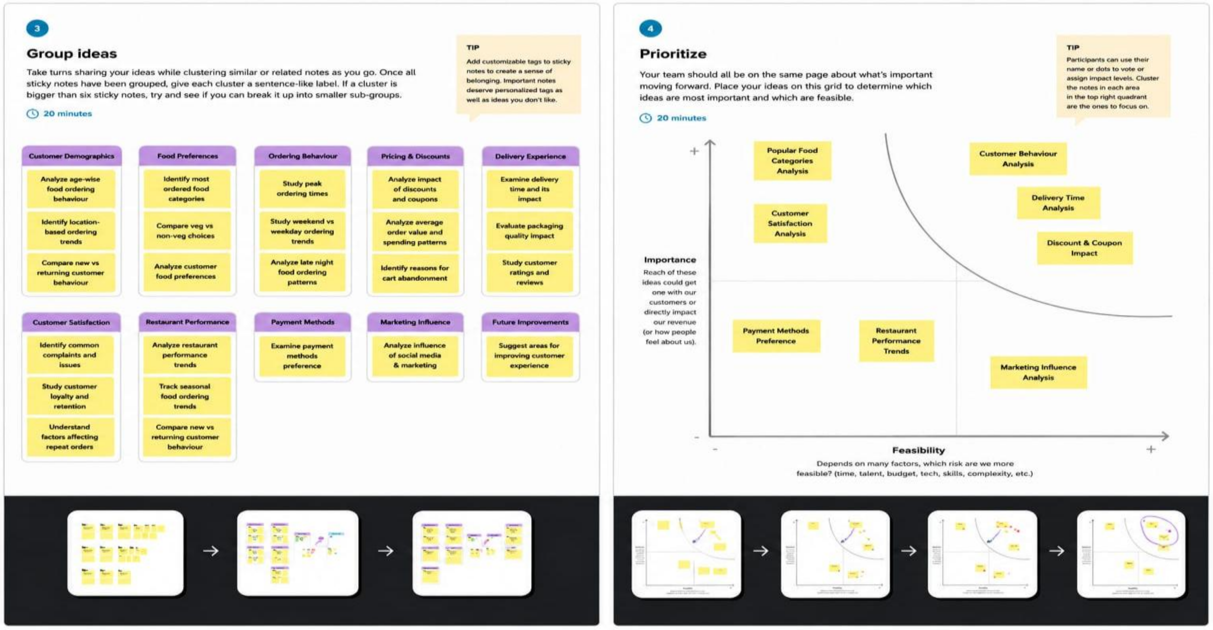
Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2: Brainstorm, Idea Listing and Grouping



Step-3: Idea Prioritization



3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

From data collection → cleansing → visualization → stakeholder review → strategic action.

3.2 Solution Requirement

- The proposed solution is designed to analyze food ordering behaviour and consumer trends using data visualization techniques. The system collects and processes food ordering data to identify customer preferences, ordering patterns, spending habits, and satisfaction levels. The insights generated help restaurants and food delivery platforms make informed business decisions..

FR No.	**Functional Requirement (Epic)**	**Sub Requirement (Story / Sub-Task)**
FR-1	User Registration	Register using Email, Mobile Number, or Google Account
FR-2	User Confirmation	Secure Login with Email/Mobile and password

FR-3	User Profile Management	Edit Personal Details ,Delivery Address, Food Preferences
FR-4	Food Search & Filter	Search by cuisine, Restaurant, price, Ratings
FR-5	Order Placement	Add to Cart, Modify Quantity, Place Order
FR-6	Payment System	UPI, Credit/Debit Card, Net Banking, Cash on Delivery
FR-7	Order Tracking	View Order Status and Delivery Updates
FR-8	Review & Rating	Rate Food, Restaurants, and Provide Feedback
FR-9	Recommendation System	Suggest Food Based on Previous Orders and Preferences
FR-10	Reports & Dashboard	View Consumer Trends, Popular Foods, Spending Analysis

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a) Functional Requirement

Following are the functional requirements of the proposed solution.

b) Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	**Non-Functional Requirement**	**Description**
NFR-1	**Usability**	Simple and User-Friendly Interface
NFR-2	**Security**	Encrypt User Data and Secure Payments

NFR-3	**Reliability**	System should be available 24×7
NFR-4	**Performance**	Pages should load within 3 seconds
NFR-5	**Availability**	The system should be available 24×7 with at least 99.9% uptime, allowing users to access food ordering services anytime with minimal downtime.
NFR-6	**Scalability**	Support a large number of users simultaneously

3.3 Data Flow Diagram

A Data Flow Diagram (DFD) is a graphical representation of how data moves through the system. It illustrates the flow of information between users, processes, and data storage. In this project, the DFD explains how food ordering data is collected, processed, analyzed, and presented through interactive Tableau dashboards.

DFD Level 0 (Context Diagram)

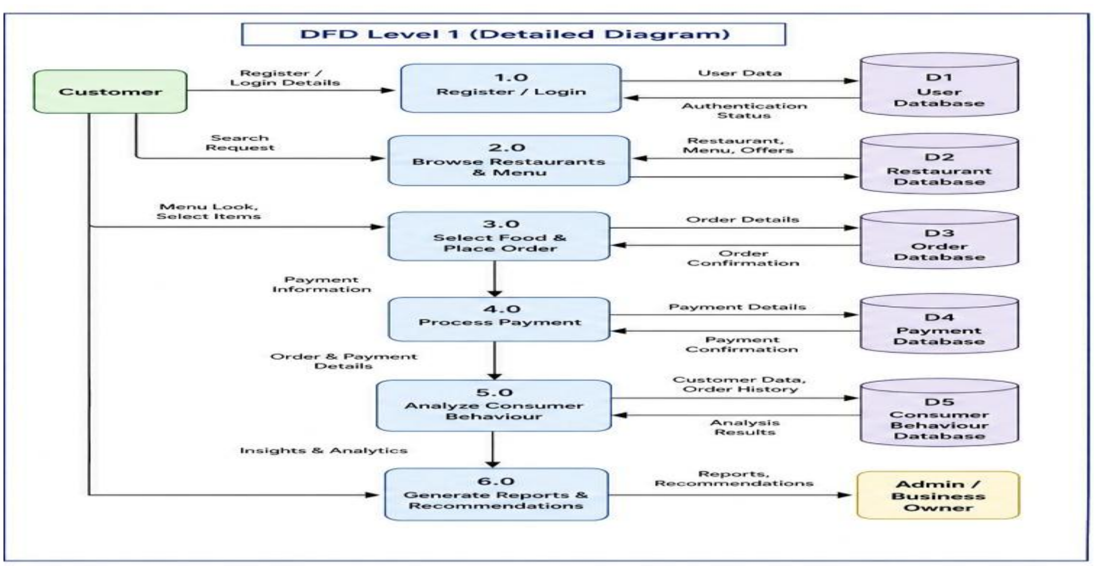
Entities:

Customer

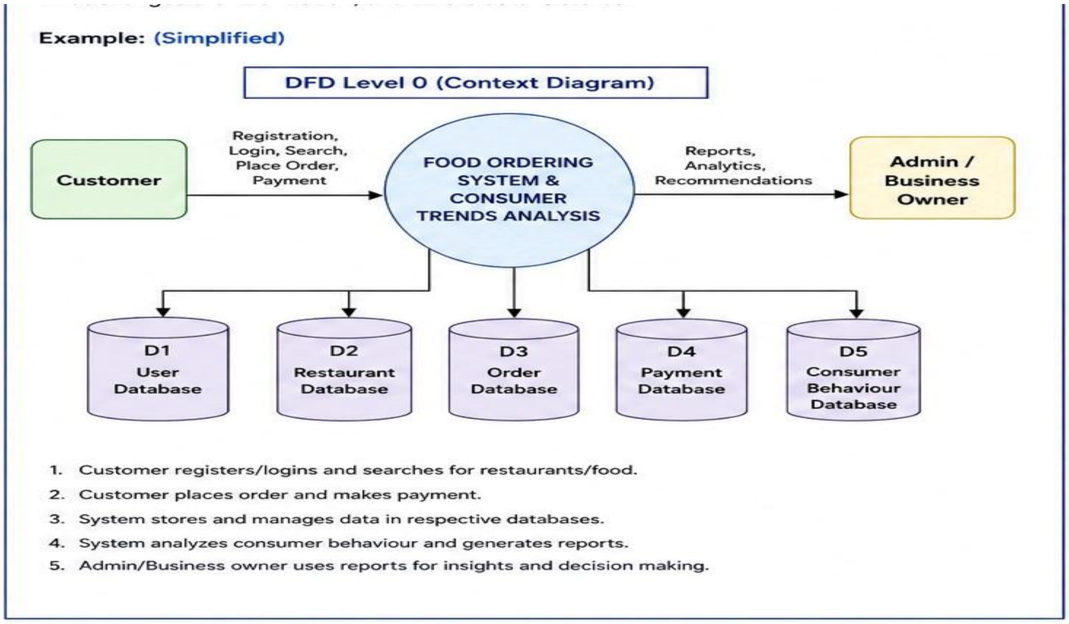
Food Ordering Dataset

Tableau Dashboard

Example:



Example: DFD Level 0 (Industry Standard)



1. User Input

Customers select restaurant, food items, and payment method.

Restaurants update menu and food details.

2. Data Ingestion

Food ordering data is imported from the dataset and stored.

3. Processing

The system analyzes customer behaviour, ordering trends, and preferences.

4. Output

Tableau dashboards display insights, trends, and reports for decision-making.

User Stories

User Stories

Use the below template to list all the user stories for the product.

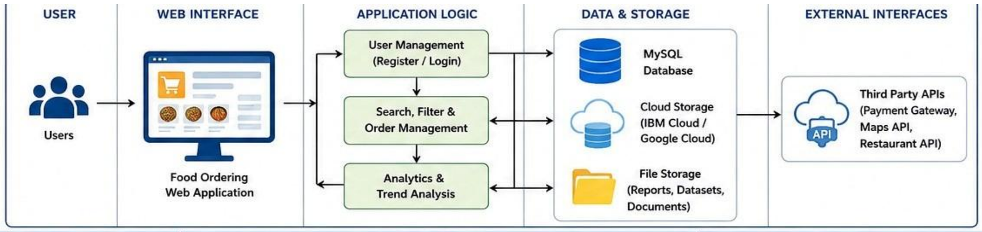
User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile User)	Registration	USN-1	As a user, I can register using my email address and password.	Account is created successfully.	High	Sprint-1
Customer (Mobile User)	Login	USN-2	As a user, I can log in using my registered email and password.	User can access the dashboard.	High	Sprint-1
Customer (Mobile User)	Browse Restaurants & Menu	USN-3	As a user, I can browse restaurants, cuisines and view menus.	Restaurants and menus are displayed successfully.	High	Sprint-1
Customer (Mobile User)	Search Food	USN-4	As a user, I can search for dishes by name, cuisine or category.	Relevant food items are displayed.	Medium	Sprint-1
Customer (Mobile User)	Place Order	USN-5	As a user, I can add food items to cart, customize and place an order.	Order is placed and confirmed successfully.	High	Sprint-1
Customer (Mobile User)	Payment	USN-6	As a user, I can make payment using UPI, Card, Net Banking or Cash on Delivery.	Payment is processed and confirmed successfully.	High	Sprint-1
Customer (Mobile User)	Track Order	USN-7	As a user, I can track my order in real time.	Real-time order status and location are updated.	High	Sprint-2
Customer (Web User)	Order History	USN-8	As a user, I can view my previous orders and details.	Order history is displayed correctly.	Medium	Sprint-2
Customer Care Executive	Customer Support	USN-9	As a support executive, I can view and resolve customer queries and complaints.	Customer issues are resolved and updated.	Medium	Sprint-2
Administrator	Analytics & Reports	USN-10	As an administrator, I can analyze consumer ordering behaviour and generate reports.	Reports and analytics are generated successfully.	High	Sprint-3

3.4 Technology Stack

- Tool: Tableau Desktop
- Data: Cleaned CSV food ordering behaviour dataset
- Environment: Local system
- Other Tools: Excel/Google Sheets (optional for preprocessing)

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits using Tableau



S.No	Component	Description	Technology
1	User Interface	Interface for users to explore restaurants, food items, place orders and view trends.	HTML, CSS, JavaScript, Bootstrap, Tableau Dashbo
2	Application Logic-1	Handles user registration, login, authentication and profile management.	Python, Flask
3	Application Logic-2	Handles food search, filtering, cart, order placement and payment process.	Python, Flask
4	Application Logic-3	Performs consumer behaviour analysis and trend identification.	Python, Pandas, NumPy
5	Database	Stores users, restaurants, food items, orders, payments and reviews.	MySQL
6	Cloud Database	Stores datasets and backups for scalability and secure access.	IBM Cloud / AWS RDS
7	File Storage	Stores datasets, reports, dashboards and project documents.	Cloud Storage / Local Storage
8	External APIs	Integrates third-party services for payments, maps and restaurant data.	Razorpay API, Google Maps API, Restaurant API
9	Analytics Engine	Analyses data and discovers consumer patterns and trends.	Python, Pandas, Scikit-learn (if applicable)
10	Reporting / Visualization	Creates dashboards and reports for insights and decision making.	Tableau

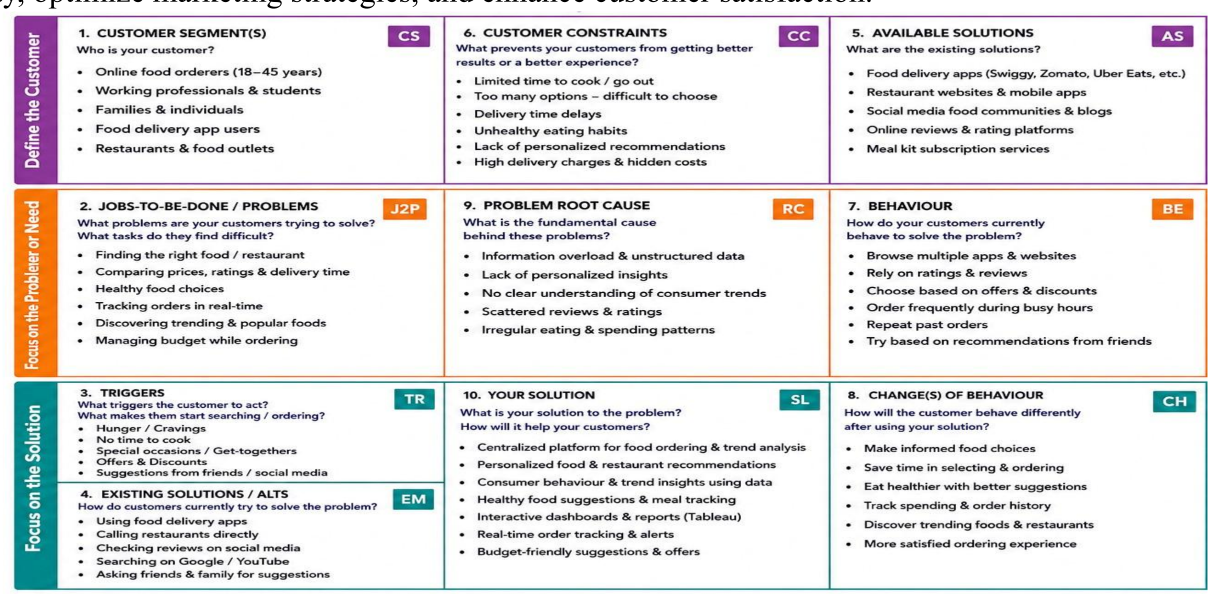
REFERENCE	Python (Flask, Pandas)	MySQL	Tableau (Visualization)	IBM Cloud	Google Maps API	Razorpay (Payment G

4. PROJECT DESIGN

4.1 Problem-Solution Fit

The increasing use of online food delivery platforms has made it difficult for businesses to understand changing customer preferences and ordering patterns. Restaurants often struggle to identify popular food items, customer spending habits, and factors affecting customer satisfaction.

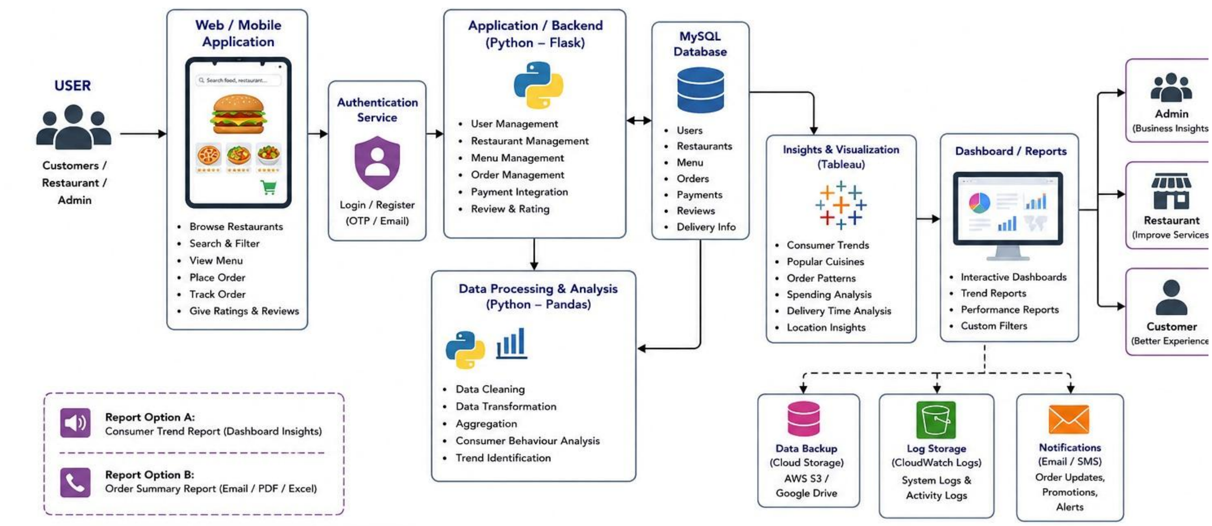
To address this problem, this project analyzes food ordering data using Tableau. Interactive dashboards visualize customer behaviour, popular cuisines, payment methods, ordering trends, and customer ratings. These insights help restaurants and food delivery platforms make data-driven decisions, improve service quality, optimize marketing strategies, and enhance customer satisfaction.



4.3 Solution Architecture

The solution architecture begins with collecting food ordering data from an Excel/CSV dataset. The data is cleaned and preprocessed to ensure accuracy. The processed data is analyzed to identify customer behaviour, ordering patterns, spending habits, and consumer trends. Interactive dashboards are created in Tableau to visualize the analysis. Finally, the dashboards generate insights and reports that help restaurants and food delivery businesses make informed decisions and improve customer satisfaction.

Example - Solution Architecture Diagram:



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Product Backlog, Sprint Schedule, and Estimation

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	User Registration	USN-1	AS a user, I can register using my email and password	2	High	Raja Rajeswari
Sprint-1	User login	USN-2	As a user, I can log in in securely to access the system	2	High	Raja Rajeswari
Sprint-1	Food Search	USN-3	As a user, I can search food by name, cuisine, or restaurant	3	Low	Raja Rajeswari
Sprint-2	Food ordering	USN-4	As a user, I can add food to the cart and place an order	5	Medium	Raja Rajeswari
Sprint-2	Payment	USN-5	As a user, I can pay using UPI, card, or cash on delivery	3	High	Raja Rajeswari
Sprint-3	Consumer Trend Analysis	USN-6	As a Admin, I can view interactive Tableau dashboards and reports	5	Medium	Raja Rajeswari
Sprint-4	Dashboard	USN-7	As an admin, I can generate consumer trend and sales reports	3	Medium	Raja Rajeswari
Sprint-4	Reports	USN-8	As a user, I can rate food items and provide reviews	2	Low	Raja Rajeswari

Project Tracker, Velocity & Burndown Chart:

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	9 June2026	15 June 2026	20	15 June 2026
Sprint-2	20	6 Days	15 June 2026	20 June 2026	20	20 June 2026
Sprint-3	20	6 Days	21 June 2026	29 June 2026	20	29 June 2026
Sprint-4	20	6 Days	30 June 2022	7 July 2026	20	7 July 2026

Velocity:

Each sprint completed 20 story points.

Total sprints = 4.

Total story points = 80.

AV=Sprint Duration/Velocity

=80/4 = 20 SP/Sprint

Average velocity = 20 SP/Sprint.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

- The Tableau dashboard loads quickly with the dataset.
- Filters and interactive charts respond smoothly.
- Data is displayed accurately without errors.
- Dashboard performance remains stable during analysis.
- Visualizations provide clear and real-time insights.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Downloaded the Data From the Kaggle From given Link
2.	Data Preprocessing	1) Removed duplicate records. 2) Handled missing values. 3) Corrected data types (Date, Price, Ratings). 4) Created calculated fields (Profit, Average Order Value, Delivery Time).
3.	Utilization of Filters	Applied filters for City, Cuisine, Restaurant, Order Date, Rating, Payment Method, and Food Category.
4.	Calculation fields Used	Created calculated fields such as Total Sales, Average Order Value, Delivery Time, Customer Rating, Profit, and Order Count.
5.	Dashboard design	No of Visualizations / Graphs – 9

6	Story Design	No of Visualizations / Graphs - 9

7. RESULTS

7.1 Output Screenshots

(Screenshots inserted in report: Dashboard Summary and Story)

Dashboard

Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits

The Food Ordering Behaviour and Consumer Trends Dashboard provides a comprehensive analysis of customer ordering patterns, meal preferences, city-wise performance, cuisine choices, weather impact, and revenue contribution. It enables users to interactively explore the data and gain valuable business insights.

1. City & Total Orders

Purpose: Displays the total number of food orders across different cities for Breakfast, Lunch, Dinner, and Snacks.
Insight Contribution: Helps identify the cities with the highest order volume and the most preferred meal times, enabling businesses to understand regional demand.

2. City vs Cuisine

Purpose: Compares cuisine preferences across different cities and analyzes repeat and non-repeat customer orders.
Insight Contribution: Identifies the most popular cuisines in each city and helps restaurants customize their menus according to local preferences.

3. Meal Time vs City

Purpose: Shows the distribution of food orders during Breakfast, Lunch, Dinner, and Snacks across cities.
Insight Contribution: Identifies peak ordering periods, allowing businesses to optimize staff, inventory, and delivery operations.

4. Meal Type vs Rainy Weather

Purpose: Analyzes the impact of rainy weather on food ordering behaviour.
Insight Contribution: Demonstrates how weather conditions influence customer ordering patterns and helps businesses prepare for seasonal demand.

5. City vs Total Orders (Pie Chart)

Purpose: Displays the percentage contribution of each city to the total number of orders.
Insight Contribution: Highlights the cities with the highest customer demand, supporting market expansion and promotional planning.

6. Meal Type vs Company

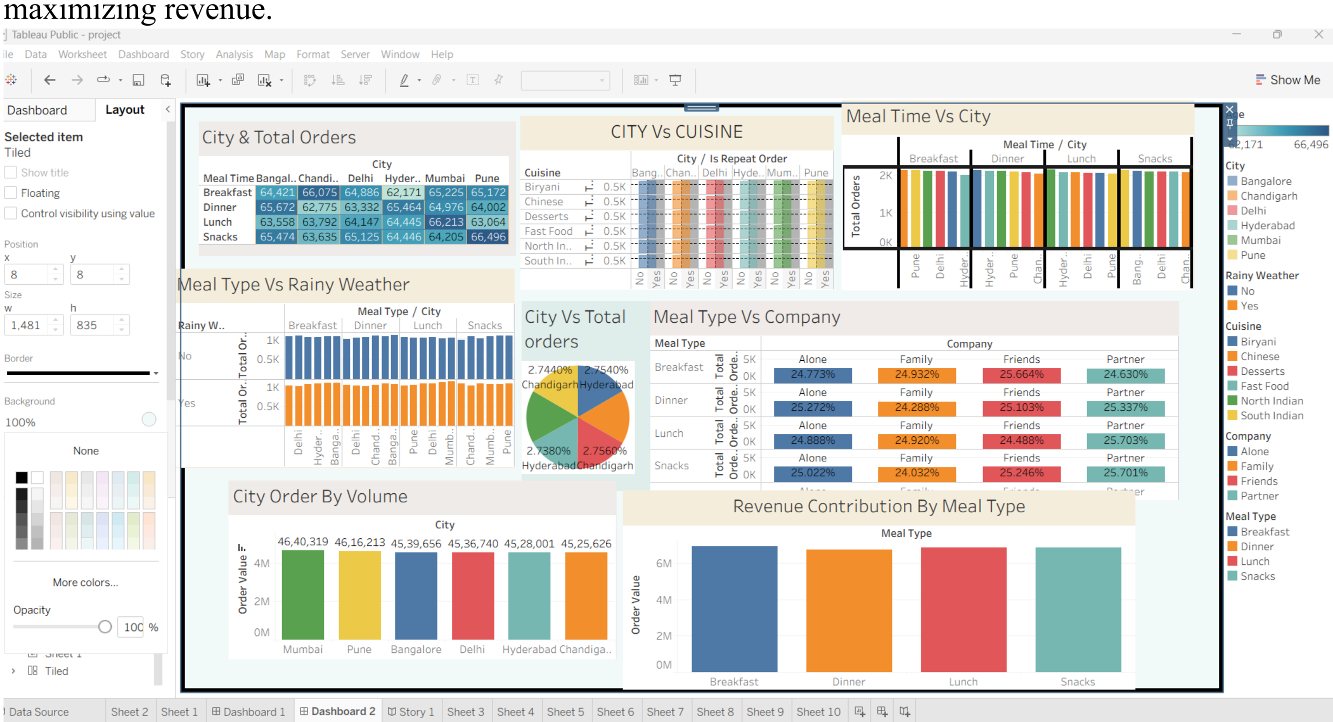
Purpose: Shows customer meal preferences based on whether they are dining Alone, with Family, Friends, or a Partner.
Insight Contribution: Helps understand customer dining habits, enabling restaurants to create suitable meal combinations and promotional offers.

7. City Order by Volume

Purpose: Compares the total order value generated by each city.
Insight Contribution: Identifies the highest revenue-generating cities and supports location-based business strategies.

8. Revenue Contribution by Meal Type

Purpose: Displays the revenue generated from Breakfast, Lunch, Dinner, and Snacks.
Insight Contribution: Identifies the most profitable meal category, helping restaurants focus on maximizing revenue.



Story

Decoding the Housing Market: A Data-Driven Journey Through Price Trends & Features

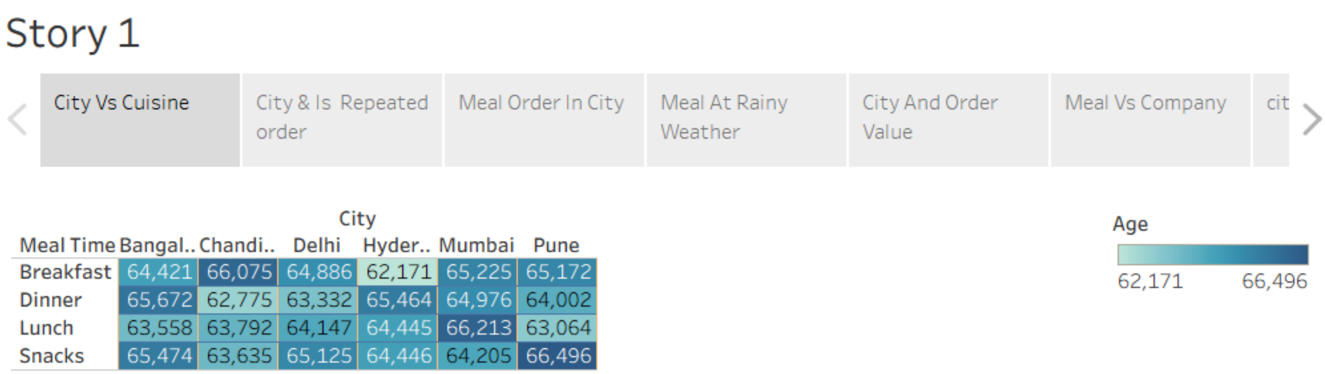
Scene 1: City vs Cuisine

1. City vs Cuisine (Repeat Orders)

Chart Type: Heat Map

Description: This heat map displays the number of repeat food orders for different cuisines across various cities. It helps compare customer preferences and loyalty towards specific cuisines.

Insight Contribution: Identifies the most preferred cuisines and cities with the highest repeat customers, supporting customer retention strategies.



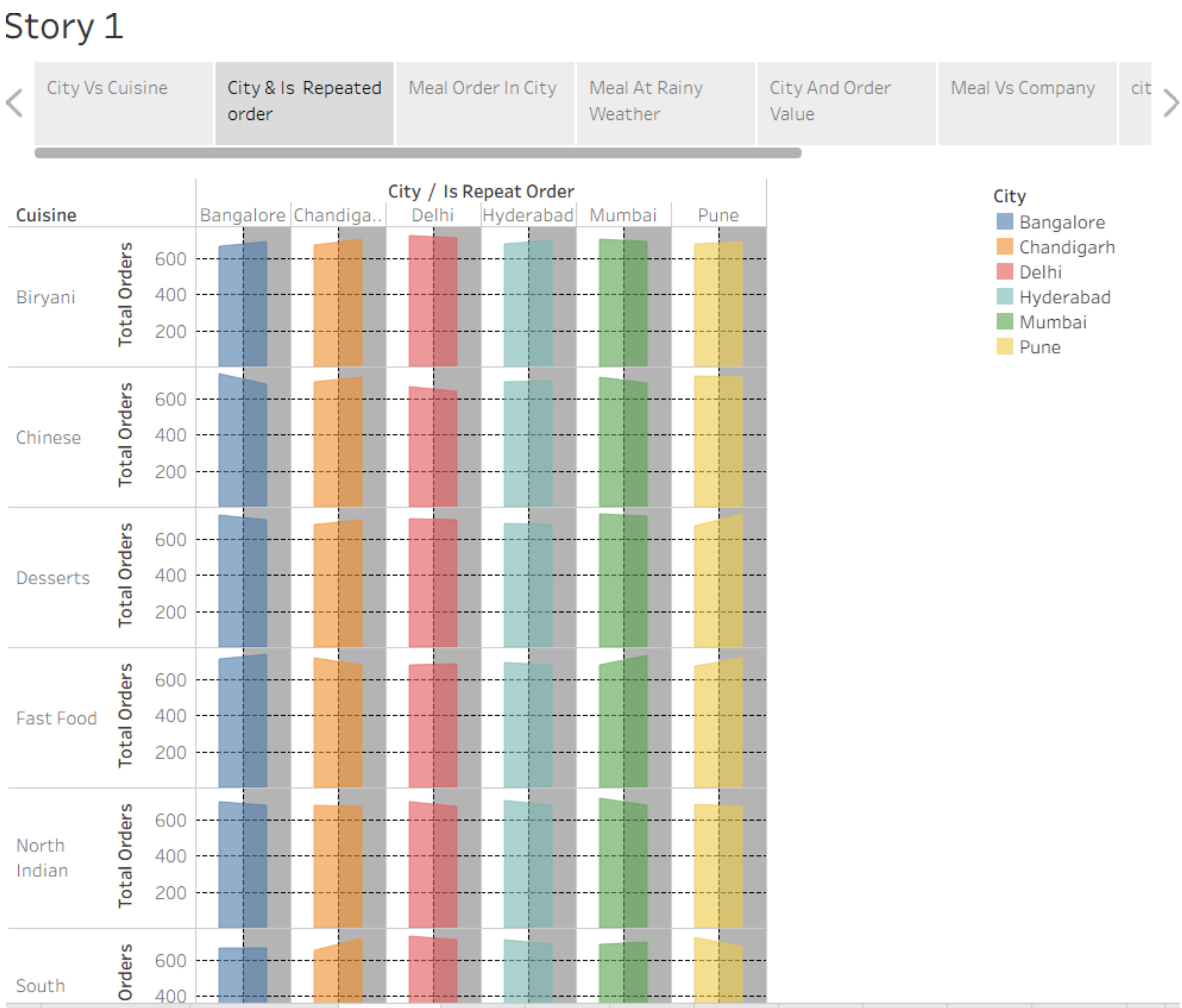
Scene 2: City & Repeated Order

2. Meal Type vs Rainy Weather

Chart Type: Bar Chart

Description: This bar chart compares the number of meal orders placed during rainy weather for different meal types such as Breakfast, Lunch, Dinner, and Snacks.

Insight Contribution: Shows how weather conditions influence customer ordering behaviour and helps businesses prepare for demand fluctuations.



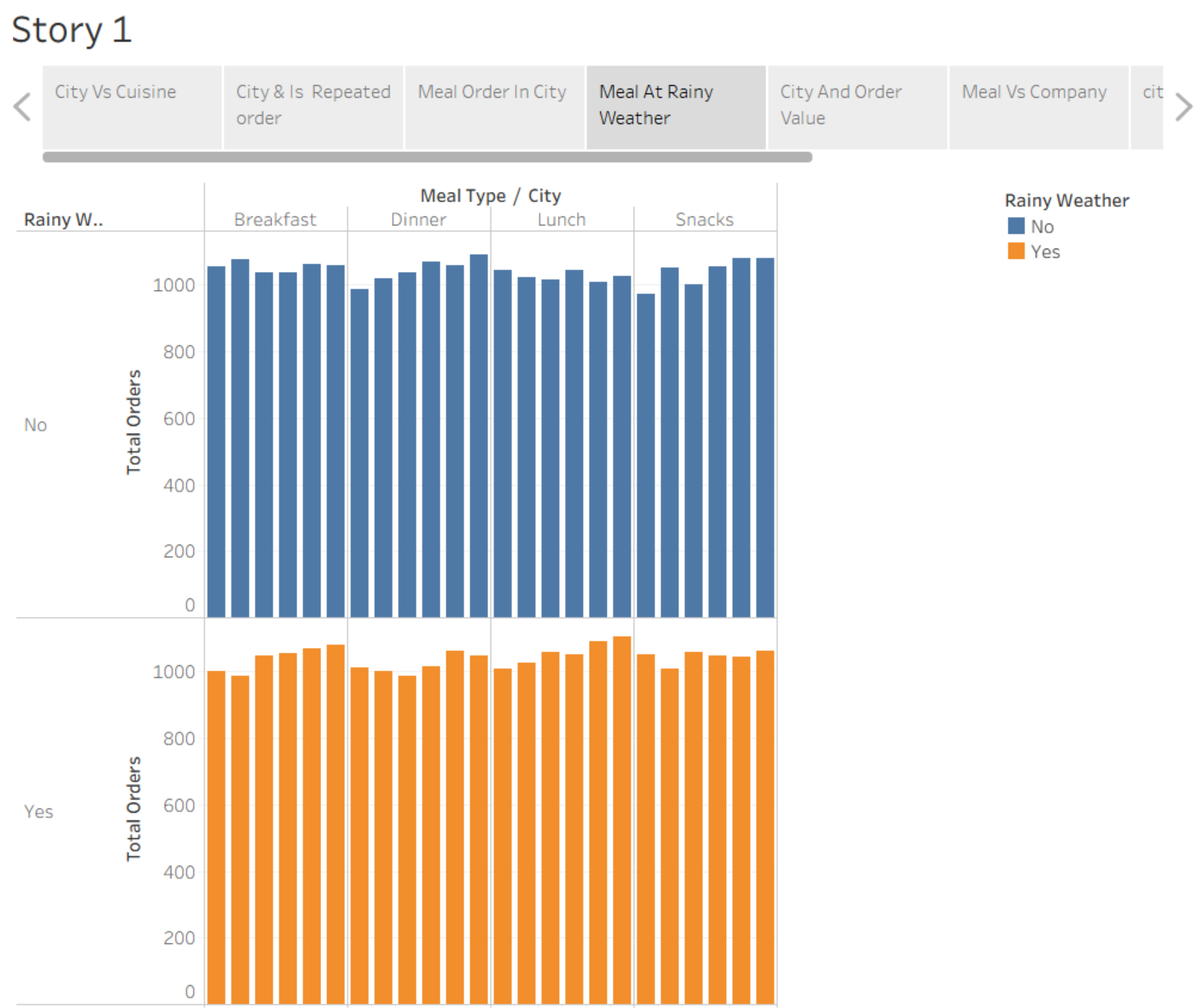
Scene 3: Meal Order in City

3. Meal Type vs Company

Chart Type: Highlight Table (Heat Map)

Description: This chart shows customer meal preferences based on their dining company, such as Alone, Family, Friends, or Partner.

Insight Contribution: Provides insights into dining habits and helps restaurants design targeted meal offers and promotions



Scene 4: Meal Type vs Rainy Weather

4. City and Order Value

Chart Type: Pie Chart

Description: This pie chart represents the percentage contribution of each city to the total order value.

Insight Contribution: Identifies the highest revenue-generating cities and supports location-based business planning

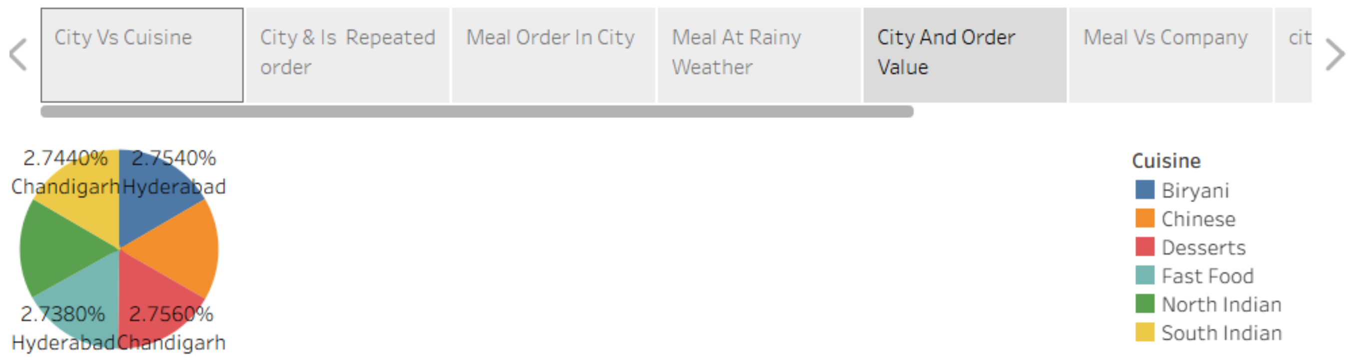
Scene 5: City And Order Value

5. City Order by Volume

Chart Type: **Vertical Bar Chart**

Description: This chart compares the total order value generated across different cities.

Insight Contribution: Highlights high-performing cities and assists businesses in focusing on markets with greater demand..



Scene 6: Meal vs Company

6. Order Time vs Time Taken

Chart Type: Line Chart

Description: This line chart illustrates the relationship between the order placement time and the delivery time taken to complete the order.

Insight Contribution: Helps identify peak ordering periods and evaluate delivery efficiency, enabling businesses to improve service quality and customer satisfaction.

Scene 7: City Order

City Order

Chart Type: Horizontal Bar Chart

Description: This horizontal bar chart displays the total number of food orders placed across different cities. It compares the order volume of each city, making it easy to identify cities with the highest and lowest customer demand.

8. ADVANTAGES & DISADVANTAGES

Advantages

Provides a clear understanding of customer food ordering behaviour.

Identifies popular cuisines, meal types, and city-wise ordering trends.

Interactive Tableau dashboards make data visualization easy to understand.

Helps businesses make data-driven decisions.

Supports better marketing strategies and customer satisfaction.

Identifies revenue-generating cities and peak ordering times.

Assists restaurants in improving operational efficiency.

Disadvantages

Results depend on the quality and accuracy of the dataset.

The analysis is limited to the available data and selected cities.

Customer preferences may change over time.

External factors such as festivals or special events are not considered.

The dashboard cannot predict future customer behaviour without predictive models.

9. Conclusion

This project successfully analyzed Food Ordering Behaviour and Consumer Trends using Tableau. The dashboard provides valuable insights into customer preferences, cuisine popularity, city-wise ordering patterns, meal timings, weather impact, and revenue generation. The analysis helps restaurants and food delivery platforms understand customer behaviour, improve service quality, optimize marketing strategies, and make informed business decisions. Overall, the project demonstrates how data visualization can transform raw data into meaningful insights that support business growth and enhance customer satisfaction.

10. Future Scope

- Integrate real-time food ordering data for live analysis.
- Apply machine learning techniques to predict customer preferences and future demand.
- Develop personalized food recommendation systems.
- Include additional factors such as festivals, holidays, and customer reviews for deeper analysis.
- Expand the analysis to more cities and countries.
- Create mobile-friendly dashboards for easier access.
- Implement AI-based forecasting to optimize inventory and delivery management.

11. Appendix :

a) Dataset Link:
https://drive.google.com/file/d/1G0gDLqog35BhOj9kuhpb5TOpKldNXYon/view?usp=drive_link

b) GitHub Link : rajarajeswarisivaprasad15/Food-Ordering-Behaviour-and-Consumer-Trends-A-Structured-Analysis-of-Choices-and-Habits

c) Video Demonstration Link:
https://drive.google.com/file/d/1NX6L45GbC-jWX2rZYZNjgqAjn0QLzOij/view?usp=drive_link